Karman Vortex Shedding over a Circular Cylinder

~23BME013

Aim: To resolve alternating vortex shedding behind a circular cylinder and estimate the Strouhal number.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cylindrical
Grid points, along radius	80
Grid points along theta	160
Reynold's number	100
Total Iterations	1200

Governing Equation:

The D2Q9 BGK lattice Boltzmann equation is:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau}\left(f_i - f_i^{eq}\right),$$

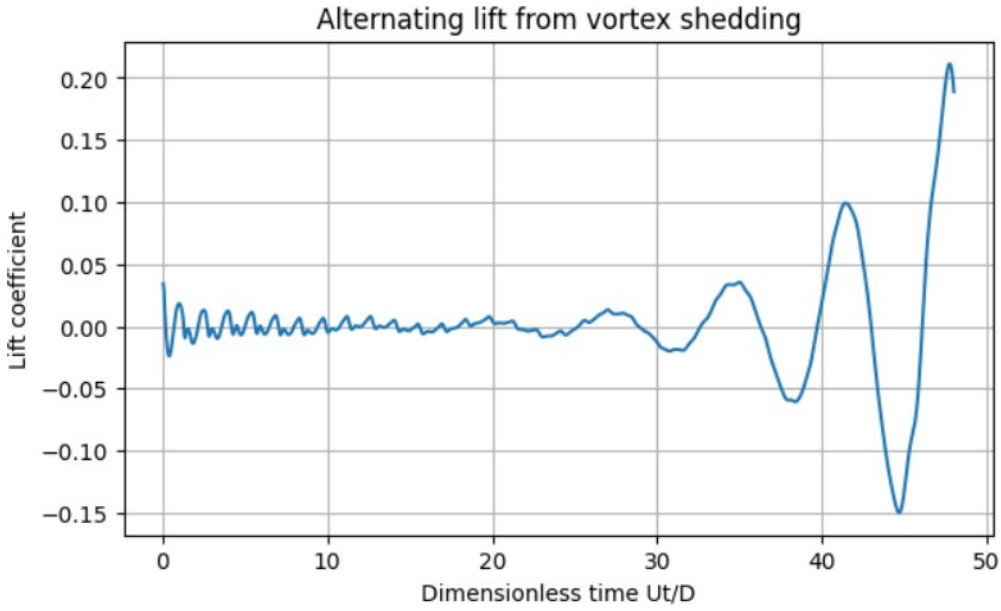
With lattice viscosity $\nu = c_s^2(\tau - 1/2)$, the shedding frequency is reported as:

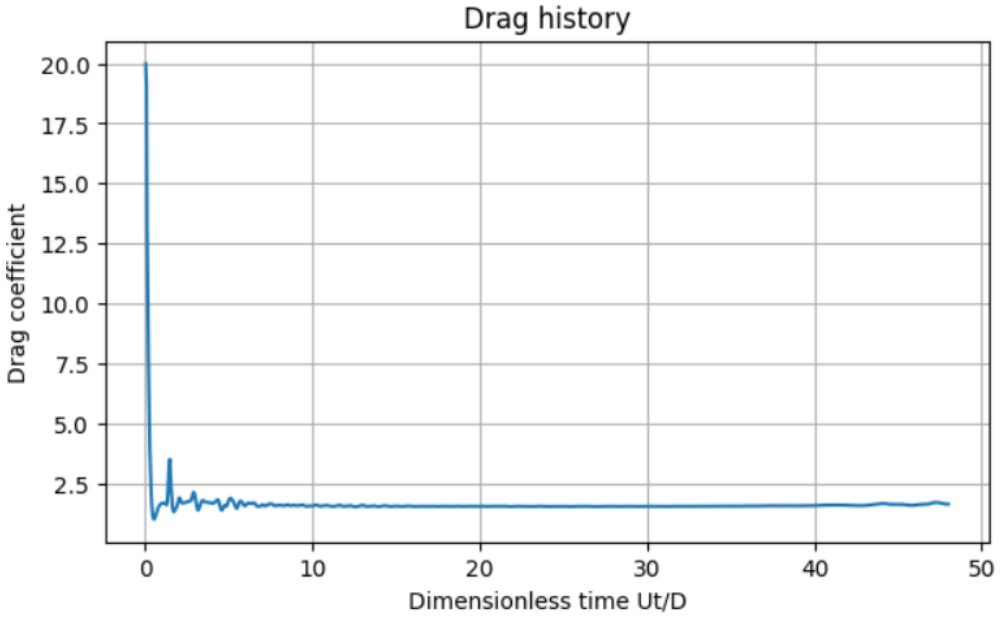
$$St = \frac{fD}{U_\infty}.$$

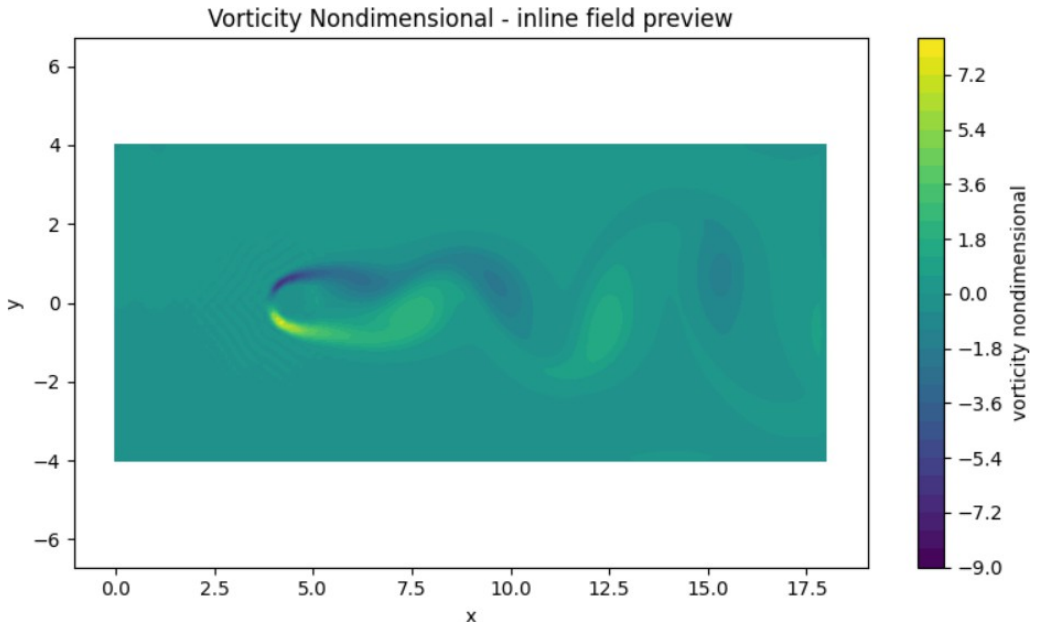
Boundary Conditions: Prescribed low-Mach inlet velocity; open outlet treatment; wall treatment on upper/lower boundaries; no-slip bounce-back at the cylinder.

Numerical Method: D2Q9 BGK lattice Boltzmann method with bounce-back cylinder treatment, momentum-exchange drag/lift evaluation, and Fourier estimation of shedding frequency.

Observations: Lift and drag histories, mean drag, lift amplitude, estimated Strouhal number, density deviation, vorticity field, and wake periodicity.


Alternating lift from vortex shedding
Lift coefficient
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
50
Dimensionless time Ut/D


Drag history
Drag coefficient
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0
10
20
30
40
50
Dimensionless time Ut/D



Results:

Parameter	**Result / Approx. Value**
Reynolds number	100
Grid	80 × 160
Total iterations	1200
Maximum positive lift coefficient,	≈ +0.21
Minimum lift coefficient,	≈ −0.15
Lift coefficient peak-to-peak	≈ 0.36
Initial maximum drag coefficient	≈ 20
Developed mean drag coefficient	≈ 1.5–1.6
Approx. vortex shedding period,	≈ 7–8
Estimated Strouhal number,	≈ 0.13–0.14
Vorticity range	≈ −9 to +7.5

Parameter	Result / Approx. Value
Wake behaviour	Alternating periodic vortex shedding

Conclusion:

The lift coefficient initially exhibits small-amplitude oscillations around zero, indicating the development of the wake behind the cylinder. As the simulation progresses, the oscillations become increasingly pronounced. Towards the end of the simulation, the lift coefficient reaches a maximum of approximately +0.21 and a minimum of approximately −0.15, giving a peak-to-peak lift variation of approximately 0.36. This alternating lift behavior is associated with the periodic formation and shedding of vortices from opposite sides of the cylinder.

The drag history shows a strong initial transient. The drag coefficient reaches approximately 20 at the beginning of the simulation and then rapidly decreases as the flow field develops. After the initial transient, C_D settles to approximately 1.5–1.6, with relatively small fluctuations about this value. This indicates that the mean drag becomes nearly steady even though the wake continues to exhibit unsteady vortex shedding.

the corresponding estimated Strouhal number from the plotted oscillations is approximately 0.13–0.14. Since the oscillations are still developing toward the end of the simulation, this should be treated as an approximate graph-based value rather than an exact numerical result.

The vorticity contour provides further evidence of the periodic wake structure. The calculated non-dimensional vorticity varies approximately from −9 to +7.5, with the highest magnitudes concentrated near the cylinder surface and in the immediate wake. Alternating positive and negative vorticity regions can be observed downstream, forming a characteristic Kármán vortex street. The vortices become progressively more diffuse as they travel downstream because of mixing and viscous diffusion.

Overall, the results show that the simulation successfully captures the principal features of unsteady flow over a circular cylinder at Re=100: an initial force transient, stabilization of the mean drag near 1.5–1.6, alternating lift fluctuations with approximately 0.36 peak-to-peak amplitude, and periodic downstream vortex shedding. These observations are consistent with the report's stated focus on lift/drag histories, mean drag, lift amplitude, Strouhal number, vorticity, and wake periodicity.